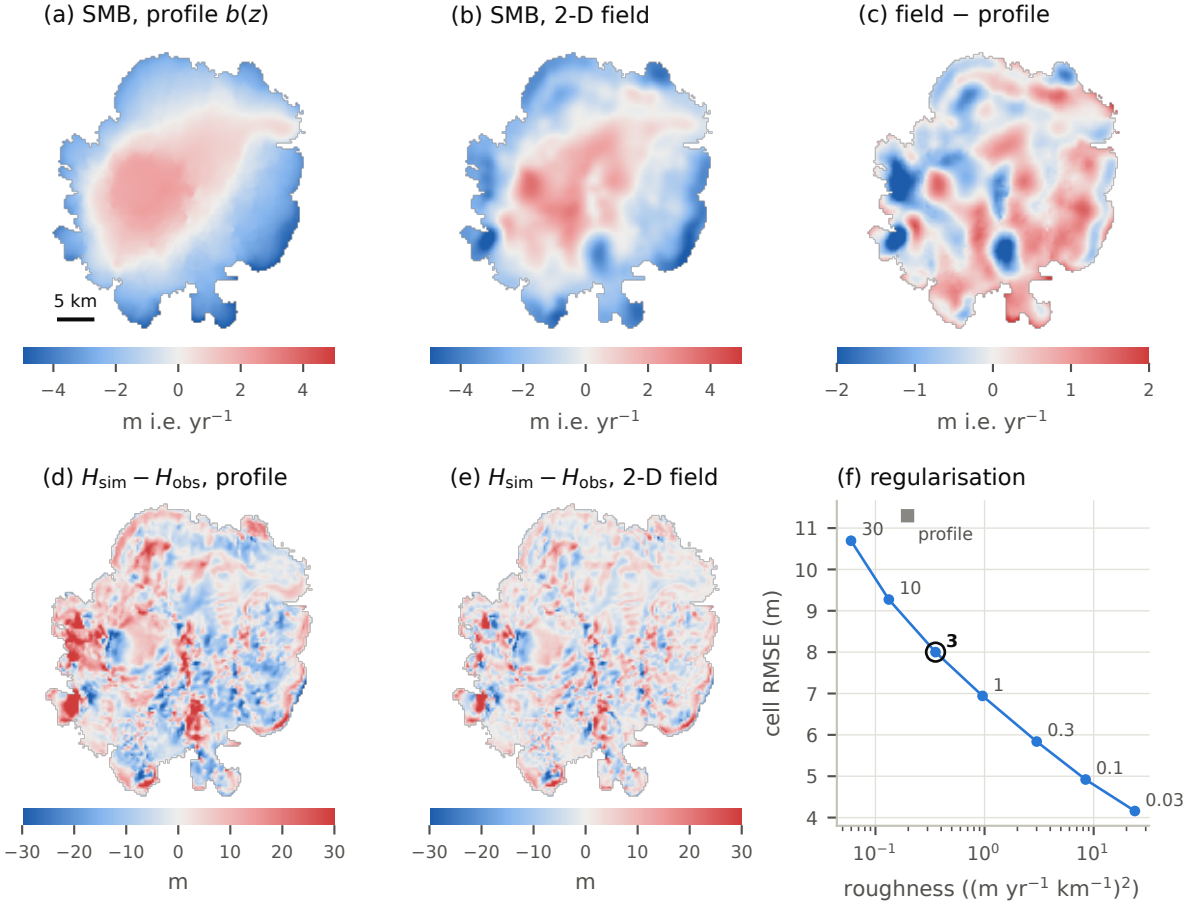

(a) SMB, profile $b(z)$
5 km
−4 −2 0 2 4
m i.e. yr^{-1}
(b) SMB, 2-D field
−4 −2 0 2 4
m i.e. yr^{-1}
(c) field − profile
−2 −1 0 1 2
m i.e. yr^{-1}
(d) $H_{sim} - H_{obs}$, profile
−30 −20 −10 0 10 20 30
m
(e) $H_{sim} - H_{obs}$, 2-D field
−30 −20 −10 0 10 20 30
m
(f) regularisation
cell RMSE (m)
11 10 9 8 7 6 5 4
30
profile
10
3
1
0.3
0.1
0.03
10^{-1} 10^{0} 10^{1}
roughness ((m yr^{-1} km^{-1})^2)